

Java String Case Study – Student Version (Questions Only)

Scenario

ABC Technologies is developing an Employee Management System. Use Java String methods to validate and process employee information. Complete the following tasks without using any third-party libraries.

Questions

1. Remove leading and trailing spaces from the employee name.
2. Convert the employee email address to lowercase.
3. Validate whether the email address is in a valid format.
4. Validate that the Employee ID follows the format EMP-1234.
5. Generate a username using the employee's first and last name.
6. Count the number of words in the employee name.
7. Determine whether the password is strong based on standard rules.
8. Mask the email address before displaying it.
9. Identify duplicate email addresses from a list of employees.
10. Extract the domain name from the email address.
11. Count the number of vowels in the employee name.
12. Count the number of uppercase, lowercase, digits, and special characters in the password.
13. Reverse the employee name.
14. Check whether the employee name is a palindrome after removing spaces and ignoring case.
15. Find the first non-repeated character in the employee name.
16. Find the most frequently occurring character in the employee name.
17. Replace all spaces in the employee name with underscores.
18. Compare two employee names using both `==` and `equals()`. Explain the difference.
19. Sort a list of employee names alphabetically.
20. Generate an employee code using the employee's initials and numeric part of the Employee ID.